LIQUID STATE MACHINES IN PARALLEL SIMULATIONS OF MAMMALIAN VISUAL SYSTEM ON RASPBERRY PI

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BACKGROUND

- Computational models of neural networks are often a primary mechanism for scientific discovery, what creates demand for larger and higher fidelity models.
- Parallel neuronal simulators have been developed that partition the neural network across multiple processing elements to speed up the computation.

BACKGROUND

- Liquid State Machine (LSM) model does not require information to be stored in stable states of a computational system, Liquid Computer is a strategy for real-time computing on time series.
- Three main components:
  - Input neurons
  - Liquid / LSM column
  - Memory-less readout map

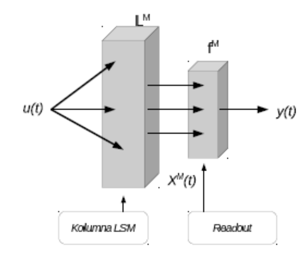
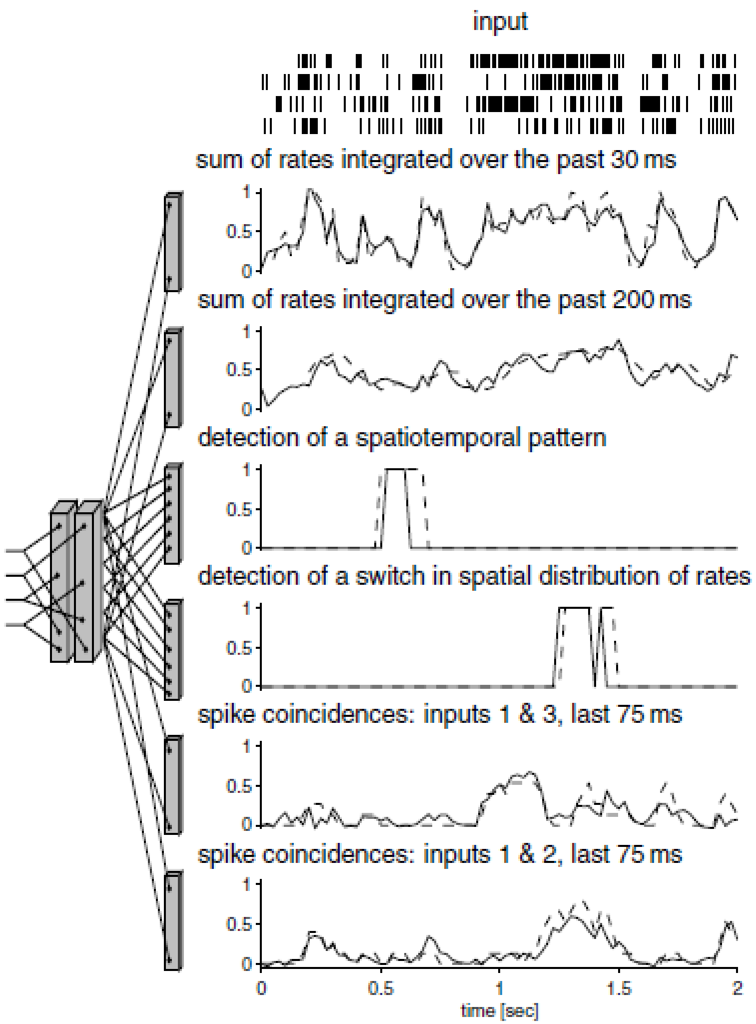


Figure: Scheme of LSM

$$X^M(t) = (L^M u)(t)$$
$$y(t) = f^M(X^M(t))$$

BACKGROUND



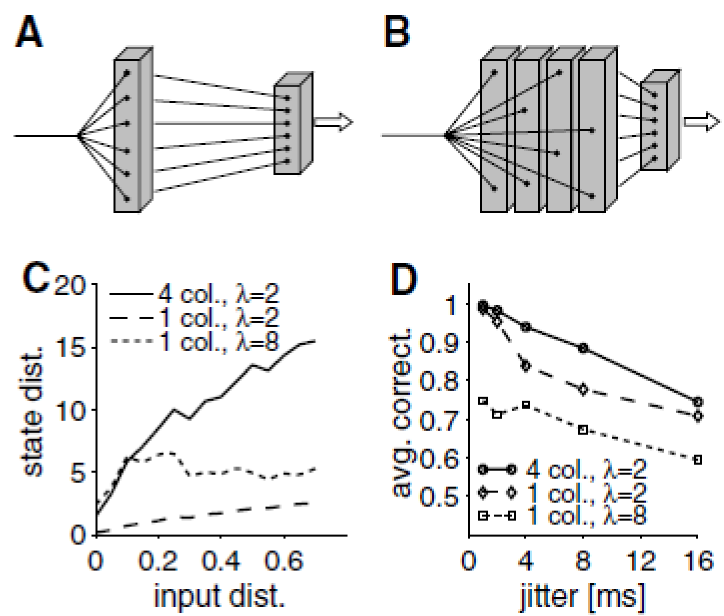


Figure 7: Separation property and performance of liquid circuits with larger numbers of connections or neurons. Schematic drawings of LSMs consisting of (A) one column and (B) four columns. Each column consists of $3 \times 3 \times 15 = 135$ I&F neurons. (C) Separation property depends on the structure of the liquid. Average state distance (at time $t = 100$ ms) calculated as described in Figure 2. A column with high internal connectivity (high λ) achieves higher separation as a single column with lower connectivity, but tends to chaotic behavior where it becomes equally sensitive to small and large input differences $d(u, v)$. On the other hand, the characteristic curve for a liquid consisting of four columns with small λ is lower for values of $d(u, v)$ lying in the range of jittered versions u and v of the same spike train pattern ($d(u, v) \leq 0.1$ for jitter ≤ 8 ms) and higher for values of $d(u, v)$ in the range typical for spike trains u and v from different classes (mean: 0.22). (D) Evaluation of the same three types of liquid circuits for noise-robust classification. Plotted is the average performance for the same task as in Figure 6, but for various values of the jitter in input spike times. Several columns (not interconnected) with low internal connectivity yield a better-performing implementation of an LSM for this computational task, as predicted by the analysis of their separation property.

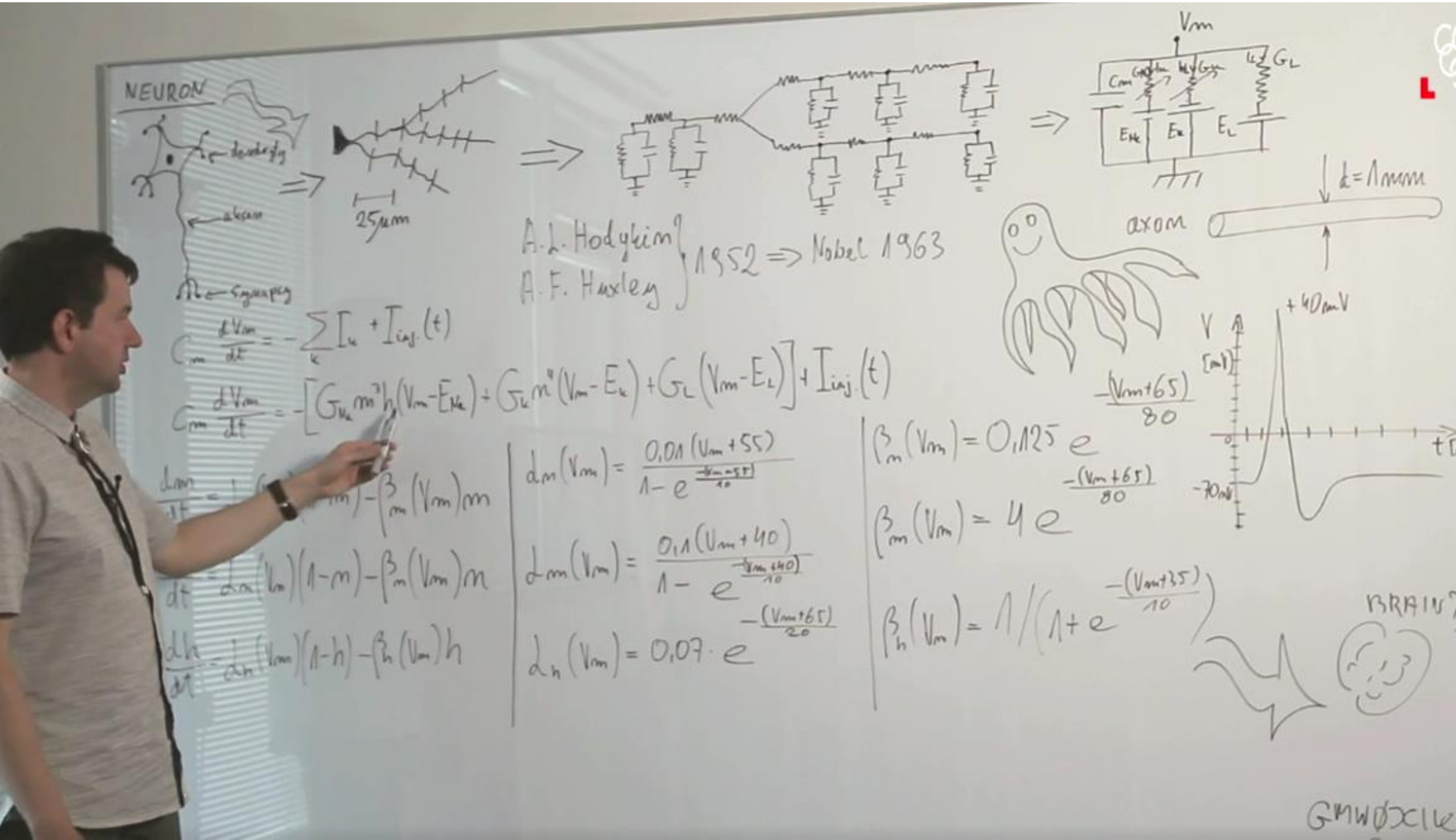
Maass, W., Natschläger, T., & Markram, H. (2002). Real-time computing without stable states: A new framework for neural computation based on perturbations. Neural computation, 14(11), 2531-2560.

STUDY: AIM

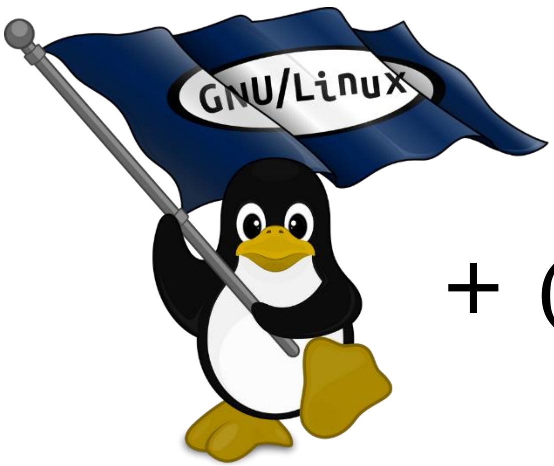
- This study examined Liquid State Machines in a simulation of a visual system on a Raspberry Pi, a tiny parallel computer.
- We expect that it is possible to design, code and runs such models for Raspberry Pi

✓ Let’s make the process of learning cybernetic modelling (and HPC) more interesting and engaging (e. g. as a new teaching aid).

METHODS: ALGORITHM

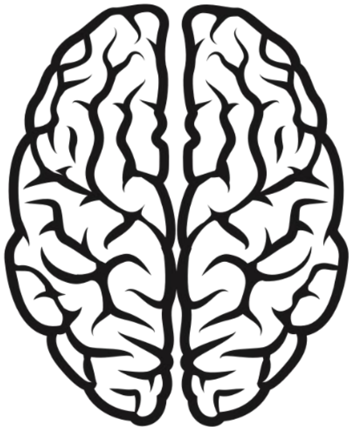


METHODS



GENESIS

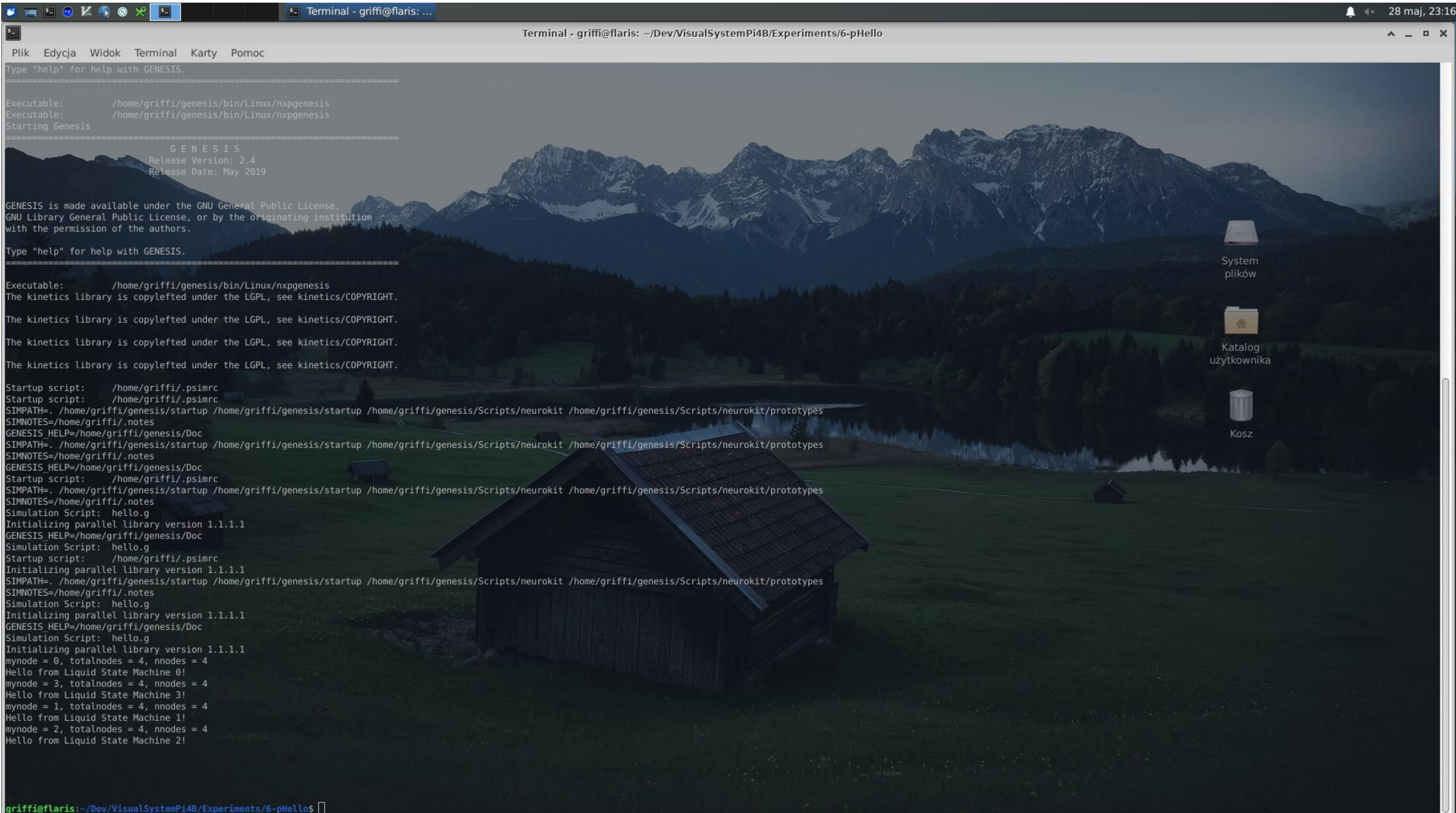
+ (the GEneral NEural =
SImulation System)



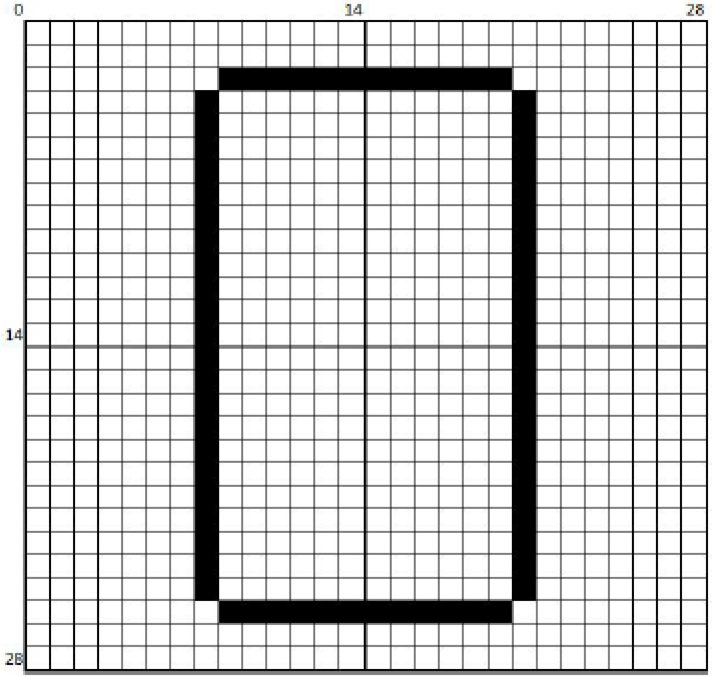
HARDWARE – RPI 4B

- **Broadcom BCM2711, Quad core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5GHz**
- **8GB LPDDR4-3200 SDRAM**
- **2.4 GHz and 5.0 GHz IEEE 802.11ac wireless, Bluetooth 5.0, BLE**
- **Gigabit Ethernet**
- 2 USB 3.0 ports; 2 USB 2.0 ports.
- Raspberry Pi standard 40 pin GPIO header (fully backwards compatible with previous boards)
- 2 × micro-HDMI ports (up to 4kp60 supported)
- 2-lane MIPI DSI display port & CSI camera port
- 4-pole stereo audio and composite video port
- H.265 (4kp60 decode), H264 (1080p60 decode, 1080p30 encode)
- OpenGL ES 3.0 graphics
- Micro-SD card slot for loading operating system and data storage
- 5V DC via USB-C connector & GPIO header (minimum 3A*), power over Ethernet (PoE) enabled. 2.5A power supply can be used if downstream USB peripherals consume less than 500mA in total
- Operating temperature: 0 – 50 degrees C ambient

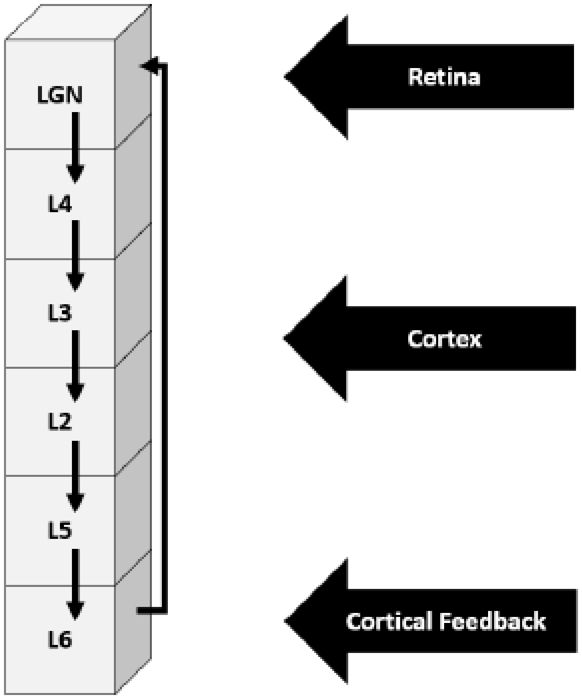
SOFTWARE


28 maj, 23:16
Terminal - griffi@flaris: ...
Terminal - griffi@flaris: ~/Dev/VisualSystemPi4B/Experiments/6-pHello
Plik Edycja Widok Terminal Karty Pomoc
Type "help" for help with GENESIS.
Executable: /home/griffi/genesis/bin/Linux/nxpgenesis
Executable: /home/griffi/genesis/bin/Linux/nxpgenesis
Starting Genesis
G E N E S I S
Release Version: 2.4
Release Date: May 2019
GENESIS is made available under the GNU General Public License,
GNU Library General Public License, or by the originating institution
with the permission of the authors.
Type "help" for help with GENESIS.
Executable: /home/griffi/genesis/bin/Linux/nxpgenesis
The kinetics library is copylefted under the LGPL, see kinetics/COPYRIGHT.
The kinetics library is copylefted under the LGPL, see kinetics/COPYRIGHT.
The kinetics library is copylefted under the LGPL, see kinetics/COPYRIGHT.
The kinetics library is copylefted under the LGPL, see kinetics/COPYRIGHT.
Startup script: /home/griffi/.psimrc
Startup script: /home/griffi/.psimrc
SIMPATH=. /home/griffi/genesis/startup /home/griffi/genesis/startup /home/griffi/genesis/Scripts/neurokit /home/griffi/genesis/Scripts/neurokit/prototypes
SIMNOTES=/home/griffi/.notes
GENESIS_HELP=/home/griffi/genesis/Doc
SIMPATH=. /home/griffi/genesis/startup /home/griffi/genesis/startup /home/griffi/genesis/Scripts/neurokit /home/griffi/genesis/Scripts/neurokit/prototypes
SIMNOTES=/home/griffi/.notes
GENESIS_HELP=/home/griffi/genesis/Doc
Startup script: /home/griffi/.psimrc
SIMPATH=. /home/griffi/genesis/startup /home/griffi/genesis/startup /home/griffi/genesis/Scripts/neurokit /home/griffi/genesis/Scripts/neurokit/prototypes
SIMNOTES=/home/griffi/.notes
Simulation Script: hello.g
Initializing parallel library version 1.1.1.1
GENESIS_HELP=/home/griffi/genesis/Doc
Simulation Script: hello.g
Startup script: /home/griffi/.psimrc
Initializing parallel library version 1.1.1.1
SIMPATH=. /home/griffi/genesis/startup /home/griffi/genesis/startup /home/griffi/genesis/Scripts/neurokit /home/griffi/genesis/Scripts/neurokit/prototypes
SIMNOTES=/home/griffi/.notes
Simulation Script: hello.g
Initializing parallel library version 1.1.1.1
GENESIS_HELP=/home/griffi/genesis/Doc
Simulation Script: hello.g
Initializing parallel library version 1.1.1.1
mynode = 0, totalnodes = 4, nnodes = 4
Hello from Liquid State Machine 0!
mynode = 3, totalnodes = 4, nnodes = 4
Hello from Liquid State Machine 3!
mynode = 1, totalnodes = 4, nnodes = 4
Hello from Liquid State Machine 1!
mynode = 2, totalnodes = 4, nnodes = 4
Hello from Liquid State Machine 2!
griffi@flaris:~/Dev/VisualSystemPi4B/Experiments/6-pHello$
System plików
Katalog użytkownika
Kosz

METHODS

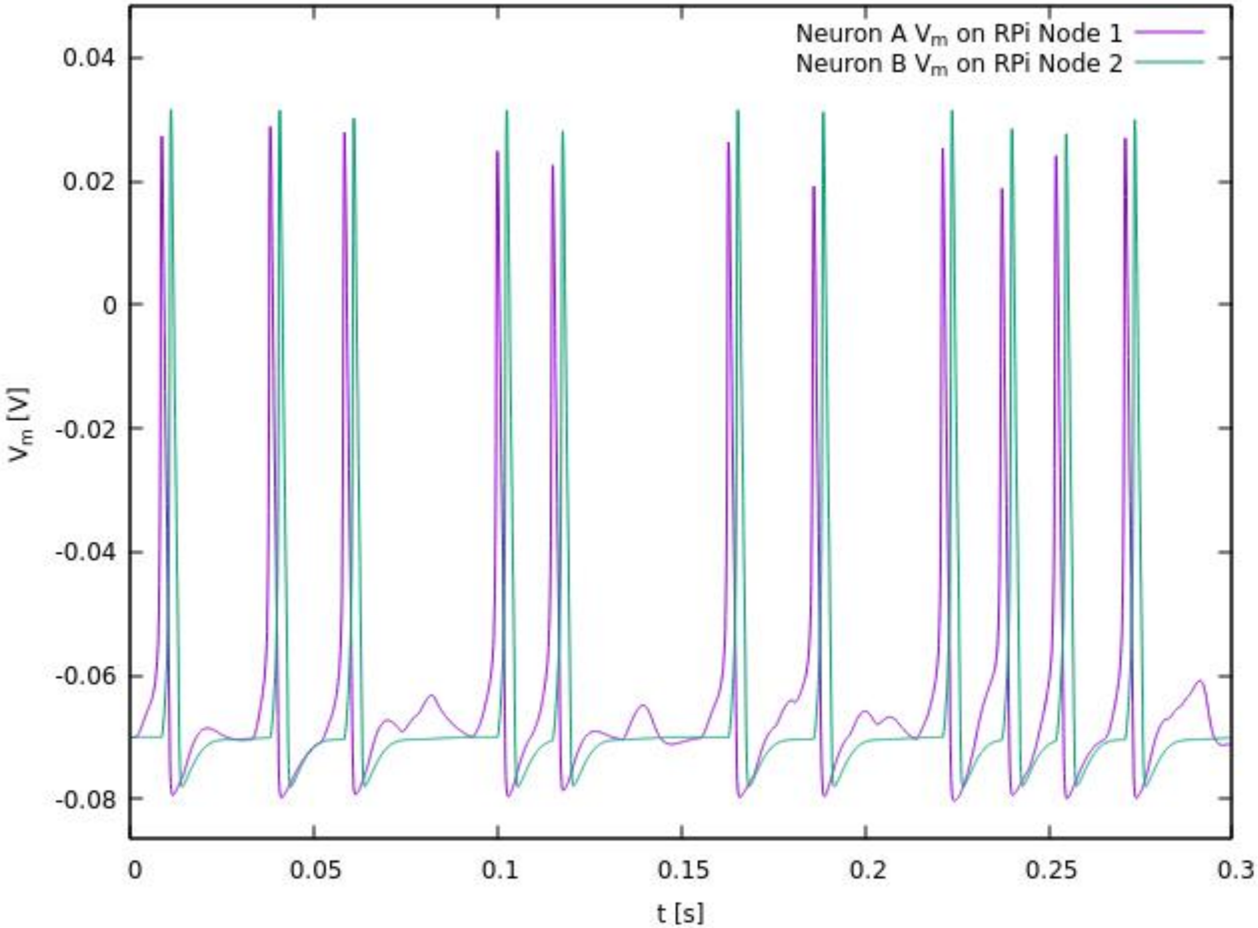


Retina of the visual system with the stimulating patterns (Input)



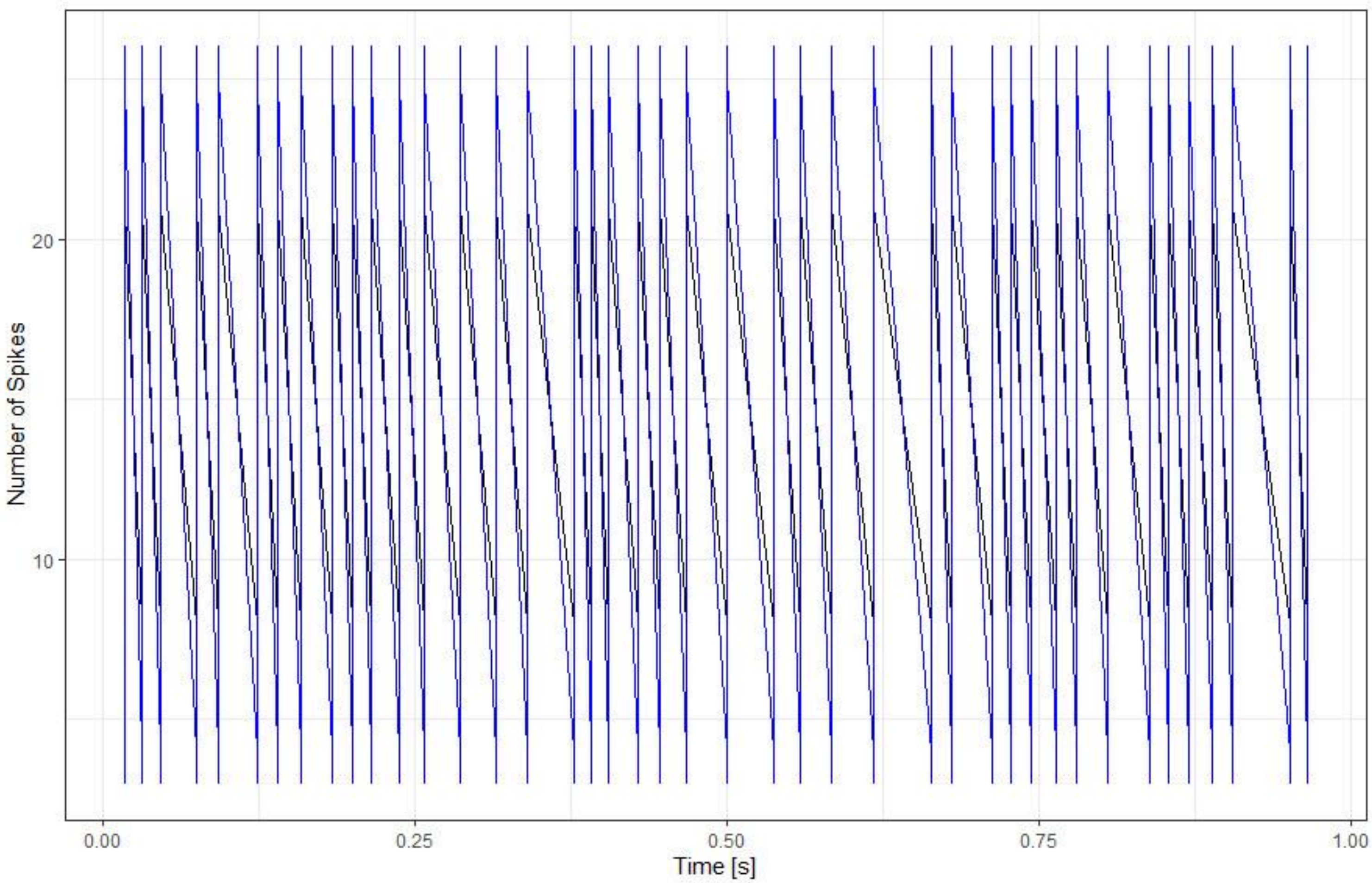
Structure of the single LSM column, a fundamental computational microcircuit of our proposed model (Liquid)

EXPERIMENTS



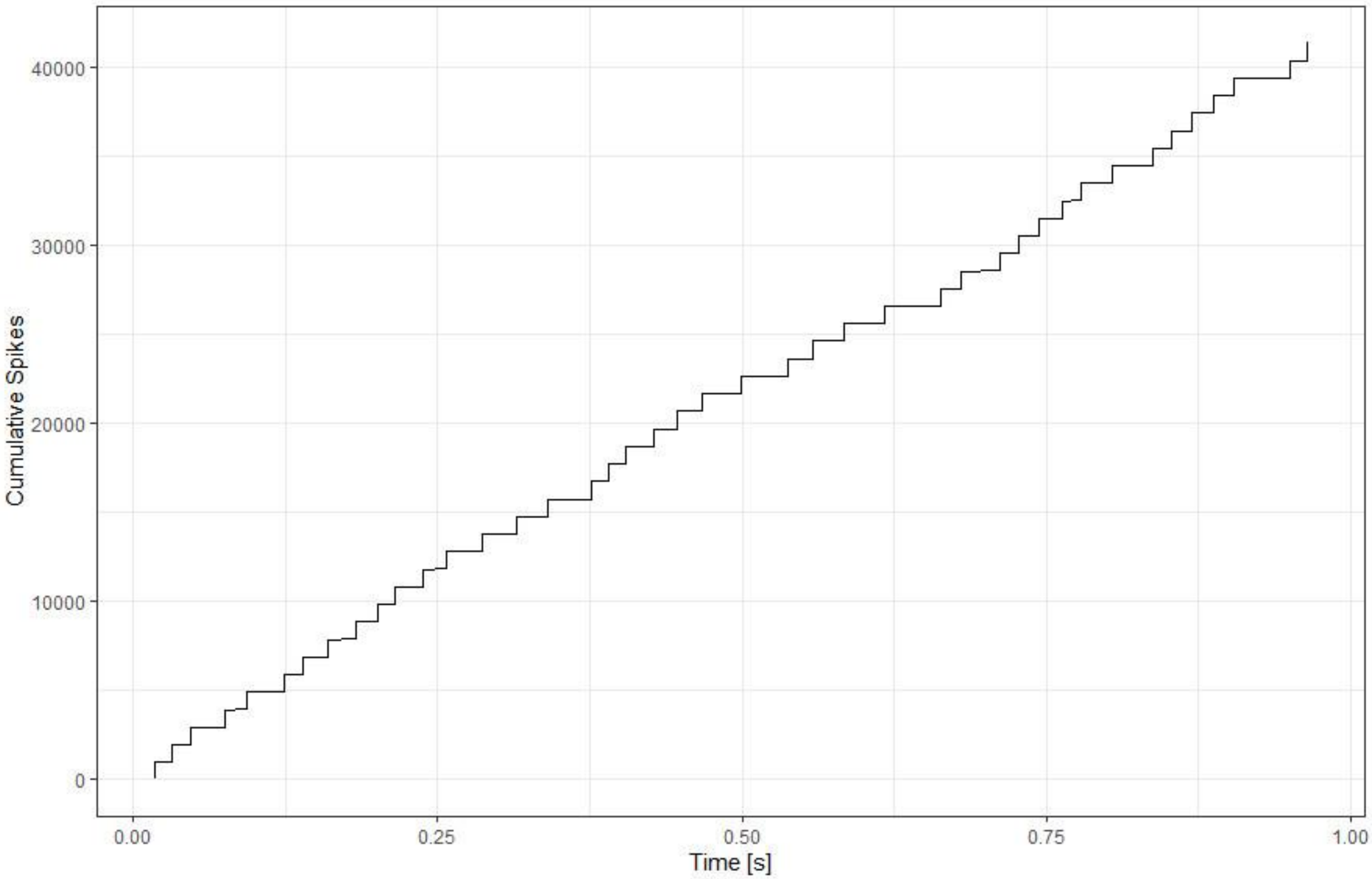
Cell membrane potential of neurons A and B in the first 300 ms

EXPERIMENTS



Retina stimulation response pattern (Number of Spikes)

EXPERIMENTS



Retina stimulation response pattern (Cumulative Spikes).

RESULTS: MAIN FINDING

- A single second our visual system built of 4880 Hodgkin-Huxley neurons was simulated (20000 steps) in 556.5 CPU seconds on RPi4B - 9:57 min

- The results could be improved with parallel optimizations or scaling up the cluster to higher node counts

RESULTS

- 2021 and Raspberry Pi 4B
  - Model 2p
    - 2 neurons
    - 20 s of biological work of the system on 3 nodes
    - 17 seconds
  - Model with 4 LSMs
    - 4x1024+768 neurons
    - 20 s of biological work of the system
    - 2.54 hours
- „19 years of simulations" (GMWójcik)
  - Model 2.5k
    - 51x51 neurons
    - 20 s of biological work of the system
      - 1998 – Sun Enterprise Ultra 3000 (the fastest computer in Eastern Poland) – 16 days
      - 2007 – laptop with core i7 - 2 hours

CONCLUSIONS

- A system based on Linux, RPi computer and GENESIS proved to be an affordable way of conducting parallel simulations of mammalian visual system.
- Can be used independently or in a cluster to simulate multiple neural columns (even parts of the brain) and make the learning of cybernetic modeling more interesting and engaging (e. g. as a teaching aid)